

Blackwater (River / Wetlands) Map Design

Executive Summary

Blackwater is a dense river/wetlands battlefield designed for water crossings, naval support, and ambush/swamp around combat. The terrain features wildflower rivers, marshes, flooded forests, unstable wetlands, elevated levees, fisher's docks, and military river outposts. The map emphasizes Outbreak Protocol's autonomous amphibious tactics, sensor-limited environments, comms-relay networks, and tactics under wetland constraints. In AR (Real Left/Map), the map anchors at ~4-5 m shallow distance, covering a 20x10 m physical area measuring ~1 km² of river territory. Key features include multi-route river crossings, swamp ambush zones, naval patrol lanes, elevated levee roads, and hidden infiltration paths.

Design Goals & Gameplay

Amphibious & Wetland Combat

Blackwater supports coordinated ground pushes, marine naval patrols, and shallow-water vehicle crossings. Marsh terrain slows movement and encourages ambushes.

Autonomy & Ambush Logic

Autonomous AI chooses between river crossings, levee roads, swamp infiltration, or dockside flanking. Units adapt to shifting water levels and visibility.

Sensor & Information Warfare

Fog, humidity, dense vegetation, and water reflections distort sensors. Relay towers on levees restore clarity. Destroying relays creates fog-of-war pockets across wetlands and river channels.

Logistics & Resources

River docks, fuel depots, and levee checkpoints serve as irregularly made. Supply convoys must navigate unstable wetland roads and exposed river crossings.

Deployment & Progress

A hard deployment budget (~10,000 points) forces players to choose between amphibious units, heavy ground armor, or stealth infantry. Blue deploys from the southwest riverbank. Red deploys from the northeast levee fortresses.

Spatial Constraints (Nreal AR)

Device Capabilities

Target laptop: "13" diagonal FHD, 1800x1080 per-eye resolution. Optimal viewing distance ~4-6 m.

Playable Area & Scaling

10x10 m physical area x 1 km² wetlands zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

100+ m physical area x 1 km² wetlands zone. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 Unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Riverbank Command) — (0,0). 140x40 m = Amphibious staging zone with patrol boats and life rafts.

Red HQ (Levee Fortress) — (1000,1000). 140x40 m fortified elevated base with radar mast and artillery.

Central Wetland Basin — (500,500). 200x200 m flooded forest, marsh roads, dense vegetation.

Fuel & Logistics Dock — (100,800). 100x100 m fuel tanks, boats, marine crates, depots.

Fishery Village Ruins — (800,300). 100x100 m collapsed huts, ovens, boats, interior routes.

Marsh River Channel — (0,400) to (400,800) flows river with tree cover and (100,400) and (100,800).

Swamp Infiltration Route — (100,0) to (400,300) dense vegetation: ideal for stealth ambushes.

Levee Road Network — (0,400) to (1000,800) elevated road with levee crosswalks.

Relay Tower Ridge — (100,800), (400,0) = Comm relay tower and sensor hub.

Naval Forward Channel — (0,800) flows river towards zone.

Chokepoints & Sightlines

River Crossings

Critical for movement; naval units can support ground forces.

Swamp Infiltration Route

Dense vegetation; perfect for ambushes and stealth.

Levee Road Network

Long-range sightlines; ideal for artillery and recon.

Fishing Village Ruins

Close-range combat with interior flanking routes.

Sightlines

Relay Tower Ridge provides long-range sensor coverage (~100 m). Fog and vegetation create fog-of-war pockets.

Deployment Zones

Blue Deployment: 100 Riverbank = amphibious staging area.

Red Deployment: 100 Levee Fortress = artillery positions.

Naval Forward Zone: Outpost at (100,800) for drone drops.

Assets & Environment

Terrain

Woods, marshes, wetlands, levees, flooded forests.

Vegetation

Dense reeds, floating trees, mangroves, grass patches.

Buildings & Props

Military: bunkers, watchtowers, radar dishes. Industrial: docks, pipelines, generators, pumps. Logistics: boats, trucks, crates, fuel drums.

Effects

Fog, humidity haze, water reflections, smoke plumes.

Audio

Water movement, insects, distant machinery, radio chatter.

LOD & Budget

Asset Cate Count

Wetland 1 200k

Marsh/River 4 80k 20k

Buildings 15 10k 70k

Roads/Lane 4 10k 10k

Vegetation 100 2k 80k

Props 50 10k 10k

Total — — ~2.5M

Texture Budget

2048 for terrain/buildings, 1024 for props, 512x1/5 compression. Target ~100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal AR Template + Nreal SDK (NREAL). ARKit/ARCore enabled.

Rendering

Forward rendering; stationary lights; baked GI; aggressive culling.

Anchor

Single spatial anchor for entire map.

Interaction

Phone gesture for pan/zoom; HUD as screen-space widgets.

Multitasker

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out river, wetlands, levees. AR anchor test.

2. Core Features — Month 2 Build basic amphibious pathfinding, autonomy behaviors.

3. AR Interaction — Month 3 Anchors, scene detection, basic interactivity.

4. Content FILL — Month 4-5 Final models, textures, lighting, audio.

5. Gameplay Polish — Month 6-7 Optimization, balance, VFX (fog, water reflections).

6. Testing & QA — Month 8 Multitasker stress test, sensor validation.

Testing Plan

Autonomy

Verify amphibious logic, swamp infiltration, levee pathfinding.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

400 FPS AR render on target device.

QA Checklist

Navigation, multi-domain coordination, deployment budget enforcement, anchor drift, sensor accuracy.